SW Engineering CSC648-848 Spring 2026

GatorBazaar

Team 07

Members	
Team Lead	Dylan Cordoni (dcordoni@sfsu.edu)
Github Maintainer	Abhay Bajaj
Back-End Lead	Sebastian Cervantez
	Ryan Zhang
Front-End Lead	Daniel De Jesus
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Milestone 1

History Table	
Date Submitted	March 18, 2026
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1. Executive Summary:

College students often have so much stuff towards the end of their four years and even semesters. These items include textbooks from past semesters, small furniture, electronics, or school supplies. At the same time, many other students are looking for affordable alternatives instead of buying new items. However, existing marketplaces such as general online classifieds are not designed specifically for university communities and can feel inconvenient or unsafe for student use. Having to meet up in locations that aren't always safe, for example Facebook marketplace, etc. GatorBazaar is a marketplace platform designed specifically for the San Francisco State University (SFSU) community where students and professors can easily buy, sell, and exchange items within a trusted campus environment.

GatorBazaar provides a simple web-based platform where users can list items they no longer need and browse items posted by others. Listings are organized into categories such as textbooks, electronics, clothing, and dorm essentials, and many other features we have coming down the pipeline, making it easy for users to quickly find what they are looking for. Students can post items for sale, upload descriptions and prices, and connect with buyers within the SFSU community. Professors can also use the platform to share or sell academic materials or recommend resources to students. By centralizing these exchanges in one place, GatorBazaar helps reduce waste, saves students money, and creates a convenient and safe way to reuse items within the campus community.

As San Francisco State University students ourselves, our team is dedicated to making a marketplace that not only our fellow students feel confident and safe using, but one that we would use as well. With student safety and convenience at our highest priority, we believe we can make GatorBazaar the ideal cornerstone for second-hand student shopping.

2. Personae:

Persona 1:

Name: John

Year: Incoming Sophomore

John, who just finished his first year of college, plans to move out of his school campus dorm and settle in an apartment near his college. He works a part-time job to pay for rent, but still receives some help from his parents, and he plans to find a roommate or two to cut the cost of rent he has to pay. His goal is to find affordable furniture near campus to decorate his apartment without spending too much and to avoid unsafe spaces. He's proficient in using web applications and knows how to navigate, but he's not experienced with buying secondhand items. He wants to buy furniture from another student near campus so he feels safe and since furniture can be expensive, buying secondhand items with the help of our application helps him find cheap options.

Persona 2:

Name: Jose Corona

Year: Freshman

Jose Corona is a new student at San Francisco State University. He is originally from San Diego, California and moved to SF to study Business Administration at SF State. As a new student at SF State, Jose lives in the dorms at SF State and is not familiar with the campus, city, and community around him. Like many students, his professors this semester have assigned him textbooks to buy for the semester. Although he has been told that he can try to download the books online, Jose prefers to buy the books and wants to find the best deals possible. Jose needs the books as soon as possible, and he doesn't want to buy the books online and believes he can find a better deal than the prices at the SF State book store. Therefore, Jose wants to browse the GatorBazaar app to look for the textbooks he needs for his classes.

Persona 3:

Name: Javier Hernandez

Year: Senior

Javier is a senior at San Francisco State studying biology. He is from Los Angeles and lives in a shared apartment near SF State. As a student who moves back to his home in Los Angeles after the semester ends, Javier has to get rid of the things he cannot carry on his flight back home. Therefore, Javier decides to use GatorBazaar to sell his items that he no longer wants. These items include items such as textbooks, clothes, electronics, appliances, class supplies and other things that are in his room he no longer wants. Javier is a very flexible man so he accepts cash or Zelle payments and has relatively low prices since his overall goal is to get rid of the items he no longer wants.

Persona 4:

Name: Khalid Kashmiri

Year: Senior

Khalid is a senior student at SFSU and is one of the admins for the GatorBazaar. He is a business student and is responsible for moderating the marketplace in order to make sure the market is safe and organized. Khalid's job is to review user reports and prevent scams. He is focused on maintaining a safe and positive market for his fellow students and has had training and previous experience in moderating people and websites in other online communities. Some of his biggest issues he encounters are spam listings and posts that violate the platform's rules, reducing others' trust in GatorBazaar. Khalid wants to make sure that the platform stays safe, positive, and has a high trust among students at SFSU.

Persona 5:

Name: Baljeet Singh

Year: Incoming Freshman

Baljeet is an International Student from India, majoring in Computer Science. Coming to a new country and getting used to the new things here is a bit challenging for him. He is planning to dorm and make some new friends. He wants to make sure he has everything ready before classes start. He moves in a few days before classes start, so he wants to purchase things that will help him get through his first semester. He wants to look into a study desk, so he can be locked in, but they all happen to be very expensive in stores and online. So now, he looks into the GatorBazaar app, trying to find himself a nice desk for the best price and also look for the required textbooks for his class.

3. High-level Use cases:

Use Case 1: Buying Furniture Using GatorBazaar

John has just completed his first year at SFSU and is moving from his dorm into an apartment near campus. Because he's working part-time and wants to minimize his expenses, he looks for affordable secondhand furniture. He uses GatorBazaar, which is limited to verified SFSU students through their university login. This gives him safety knowing that he would only be meeting up with other students and not strangers. He filters listings to show items he needs like tables, bed frames, chairs, and shelves so everything he needs is organized. Since John is not experienced with buying secondhand furniture, knowing the seller is an SFSU student and meeting in campus locations or nearby makes him feel safer. The application allows him to furnish his apartment affordably while maintaining a sense of security within the SFSU community.

Use Case 2: Buying Textbooks Using GatorBazaar

Jose's classes require him to get those textbooks, so he decides to use GatorBazaar to search for the books at a good price. By searching the listing tags for the courses he's taking, he's able to easily find listings specific to those courses. This way, not only can he find the required textbooks more easily, but he can make sure that they're the ones he needs for his courses. Additionally, since the app can only be used by SF State students and faculty, he can assure that the materials are those used for SF State classes and that the person selling them can authenticate the use of the textbook in the class. Also, since Jose is unfamiliar with San Francisco, he prefers to use the app since he can choose to meet up at a location on campus that will facilitate purchasing items.

Use Case 3: Selling Items Using GatorBazaar

Javier has to move out of his apartment by the end of the semester he uses Gator Bazaar to sell the stuff in his apartment that he currently does not want. Since the app is only for SF State students and faculty he knows that he can target and sell to students who might want to buy his items. Javier can log into the app and post the items he wants to sell such as a chair and desk and list them under furniture so that other students can buy his items.

Use Case 4: Finding a Roommate Through GatorBazaar

Jose is finishing his first year of college and plans to move out of the dorms the following semester, he wants to find a roommate to help reduce the cost of rent. He doesn't want to live too far out of campus and with non-students, so he uses GatorBazaar to connect with other verified SFSU students who are also looking for housing. By using the app, he can search for listings from students who are looking for roommates in nearby apartments. Since the platform is exclusive to SF students, Jose feels more comfortable communicating and arranging meetings on or near campus.

Use Case 5: Moderating the Platform with Admin Tools.

Khalid is spending some time reviewing listings posted by users on GatorBazaar in order to ensure that they follow the platform's rules and guidelines in order to keep SFSU students safe and their trust in the platform high. When Khalid finds a listing that violates the rules, or sees a listing that was reported he will remove the listing that is active, and if the SFSU user who posted the listing has a low rating score, Khalid may suspend that reported users account based on previous violations.

4. List of main data items and entities – data glossary/description:

- **Admin Users** - Users that manage reports received by either listings or users. Able to issue warnings to users about violations or take action in the form of removing a listing or banning a registered user. Reports will be displayed on the Registered Users profile or Listing Page.
- **Unregistered Users** - Users who can only access the homepage, browse, search, and filter listings, and view listing details. Do not need to log in or register.
- **Registered Users** - Users that have signed up on the site using a SFSU verified email.
    - **Buyers** - Registered users looking to purchase listings on the site. All registered users are treated as buyers innately.
        - **Sent Request** - Used to show a buyer's interest in a listing that is sent to the seller.
        - **Sent Request List** - A list showing all items a buyer has sent a request to buy (seen by buyer only).
        - **Bought Listings List** - A list of listings previously purchased by a buyer.
    - **Sellers** - Users that have at least one listing on the site.
        - **Listing -** An item for sale posted by and belonging to a registered user.
            - **Title -** The name of the item for sale.
            - **Image** - An image of the item for sale set by the seller.
            - **Description -** A description of the item for sale set by the seller.
            - **Price** - Price of the item set by the user.
            - **Category** - Grouping of a listing provided by the site.
            - **Tags** - Additional searchable terms set by the seller.
                - **Course** - Sellers can also set an SFSU course as a tag for listings related to a course.
            - **Pending Request** - A request received by the seller used to show a buyer's interest in the listing.
            - **Listing Report** - A report specific to a listing sent by other users if the listing breaks the guidelines of the site (seen by admin)
            - **Active -** Shows whether listing is currently being sold or not (system purposes)
        - **Active Listing List** - A list of all listings a seller currently has on the user's page (seen by all users)
        - **Sold Listing List** - A list of all past listings sold by a seller on the user's page (seen by user owning the list and admins)
    - **Messaging** - Allows communication between the seller and the buyer.
        - **Message** - A communication instance between registered users.

        - **Inbox** - A list of current and past messages between registered users. Allows registered users to send messages to other registered users.
        - **Notification** - When a registered user receives a message from another registered user, their inbox will indicate that they have a new message.
    - **Meet up** - A tracked location on campus where the buyer and seller agree to enact the exchange.
    - **User Rating** - Aggregate rating given to a seller/buyer by other users (seen by all users)
    - **User Report** - Report specific to a registered user sent by other registered users if guidelines of the site are broken by the reported user. (seen by admins)
    - **Email Verification** - Checks whether a user attempting to register has a valid SFSU email.
- **Search** - A method of filtering listings by their attributes(name, tags, price, etc.)
    - **Roommate Search** - Searching listings within the apartment category to find roommates.

5. Functional Requirements

Unregistered Users (5 Reqs)

1. The system shall allow visitors to access the marketplace homepage.
2. The system shall allow visitors to browse publicly available item listings.
3. The system shall allow visitors to search listings using keywords.
4. The system shall allow visitors to filter listings by category.
5. The system shall allow visitors to view listing details including title, description, price, and images.

Registered Users (11 Reqs)

6. The system shall allow users to register using an SFSU email address.
7. The system shall verify that registered users are associated with an SFSU email domain.
8. The system shall allow users to receive notifications when they receive messages.
9. The system shall allow users to report listings that violate platform rules.
10. The system shall allow users to give a rating to either seller or buyer.

Seller Specific:

11. The system shall allow registered users to create listings for items they wish to sell.
12. The system shall allow users to upload images when creating listings.
13. The system shall allow users to list a filter to their listing such as course-specific listings, furnitures, or rooms.
14. The system shall allow users to edit their listings.

15. The system shall allow users to delete their listings.
16. The system shall allow users to mark listings as sold.

Buyer Specific:

17. The system shall allow users to initiate the sending of messages to other users regarding listings.
18. The system shall allow users to search with word-specific filters to browse for listings.

Administrative Users (5 Reqs)

19. The system shall allow administrators to review reported listings.
20. The system shall allow administrators to remove listings that violate policies.
21. The system shall allow administrators to suspend users who repeatedly violate rules.
22. The system shall allow administrators to monitor marketplace activity.
23. The system shall allow administrators to manage listing categories.

6. Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and cloud servers approved by Class CTO and as agreed in M0
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All our selected application functions shall be rendered well on mobile devices (no native app to be developed)
4. Posting sales information and messaging to sellers shall be limited only to SFSU students
5. Critical data shall be stored in the database on the team's deployment server.
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application shall follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used

13. No e-mail clients or chat services shall be allowed. Interested users can only message to sellers via in-site messaging. One round of messaging (from user to seller) is enough for this application
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Media formats shall be standard as used in the market today
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "SFSU Software Engineering Project CSC 648-848, Spring 2026. For Demonstration Only" at the top of the WWW page Nav bar. (Important to not confuse this with a real application). You must use this exact text without any editing.

7. Competitive Analysis:

Feature	Facebook Marketplace	Craigslist	OfferUp	Mercari	(Our App) GatorBazaar
Verified/Secure Identity Accounts	X	X	✓	✓	✓✓✓
Message between Buyer & Seller	✓✓✓	✓	✓	✓✓✓	✓
Supports In-person Meetups/Exchanges	✓	✓	✓	X	✓✓✓
Campus-specific Users & Features	X	X	X	X	✓✓✓

Browse & Search Listings	✓✓✓	✓✓✓	✓✓✓	✓✓✓	✓
In-app Purchases/Payment	**X**	**X**	✓	✓	**X**
Roommate finding	✓	✓	**X**	**X**	✓✓✓

X = Does not support function, ✓ = Does support function, ✓✓✓ Does support and is superior

Existing online marketplaces such as Facebook Marketplace, Craigslist, OfferUp, and Mercari allow users to buy and sell items locally. These platforms provide common marketplace features such as browsing listings, messaging between buyers and sellers, and posting items for sale. However, these platforms are designed for the general public, meaning users often interact with strangers and may need to travel significant distances to complete exchanges. Our proposed SFSU Marketplace, GatorBazaar, focuses specifically on the SFSU student community market. By requiring verification through an SFSU email address, the platform creates a more trusted environment where users know they are interacting with fellow SFSU students and faculty. In addition, the system prioritizes convenient on-campus meetups rather than shipping or online payments, making exchanges simpler and safer for students. This focus on a verified campus community provides a targeted advantage over existing public marketplace platforms for common products that users may want to exchange in a close, local environment.

8. High-level system architecture and technologies used:

Software stack:

- Server Host: AWS EC2 t3.small
- Operating System: Ubuntu 24.04.1
- Database: MySQL v 8.0.45
- Web Server: Apache 2.4.58
- Server-Side Language: JavaScript

Additional Technologies:

- Web Framework:
  - Express.js v 5.2.1
  - Node.js v 22.12.0
- Web Analytics: Google Analytics

SSL Cert: Lets Encrypt (Cert Bot)
React: 19.2.0

Browsers Supported:

Google Chrome version 142 and later
FireFox version 146 and later

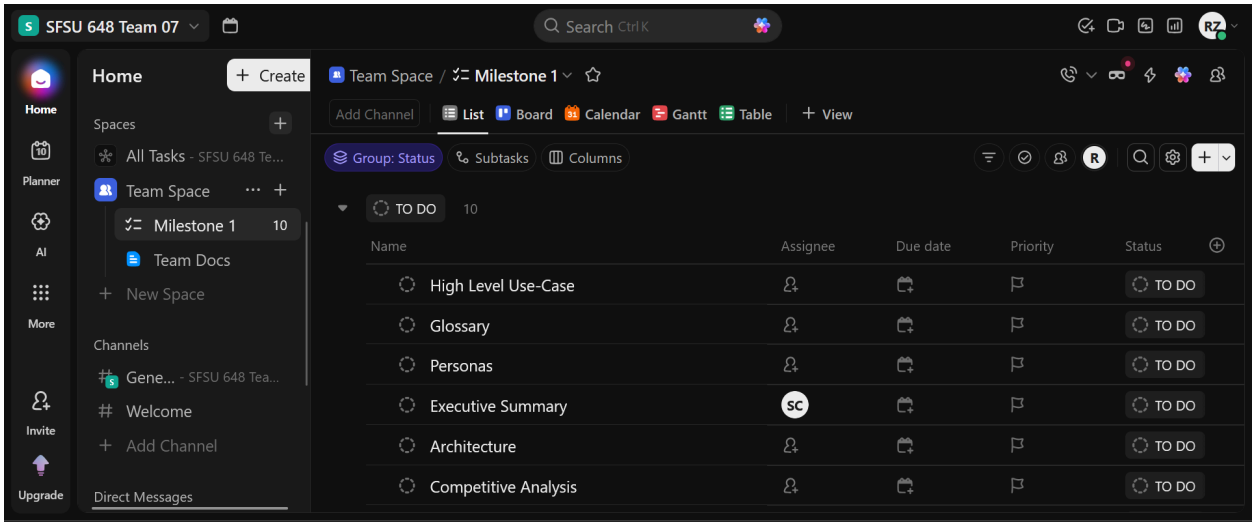
9. Use of GenAI tools like ChatGPT and Copilot for Milestone 1:

We used ChatGpt 5.2 in the process of working on M1 for tasks like:

- Deciding on a project name: Medium
    - We used the prompt “come up with a name for my website which is a marketplace for SFSU students and faculty” to get suggestions for our project name, since we had difficulty coming up with our own. From these suggestions we voted on a specific name to use. While it would’ve been easy to come up with our own names, we liked the names it suggested and it helped us reduce this process to only a few minutes.
- Summarize/condense a draft for the Competitive Analysis: High
    - We used the following prompt, “Summarize in one paragraph what are the planned advantages or competitive relationship of your planned product to what is already available. In the table clearly mark your product, e.g. shade its column/data. Your SFSU specific functions must also be addressed in competitive analysis and shown in the list of compared functionalities. I already have a table, and a competitive analysis, but I wrote it too long and I think it needs some revision to fit with the requirements listed above. Below is my table and summary.” for this task. The output for the prompt was good, and was used as a basis of our final Competitive Analysis summary, but still needed some tweaks to the grammar and to fit our purposes better.

10. Task Management Tool Selection:

Task Management Tool: Click Up



11. Team and Roles:

Roles Table		
Name	Email	Role
Dylan Cordoni	dcordoni@sfsu.edu	Team Lead
Daniel De Jesus	jdejesus1@sfsu.edu	Front-End Lead

Roles Table		
Ryan Zhang	rzhang13@sfsu.edu	Back-End
Kenny Camarra	kcamarra@sfsu.edu	Front-End
Abhay Bajaj	abajaj@sfsu.edu	Github Maintainer/Front-End
Diego Rivera-Soto	driverasoto@sfsu.edu	Front-End
Sebastian Cervantez	scervantez@mail.sfsu.edu	Back-End Lead

12. Team Lead Check List:

Check List	
Done	All team members are fully engaged and attend team sessions when required
Done	Team found a time slot to meet outside of class
Done	Team ready and able to use the chosen back and front-end framework and those who need to learn are working on learning and practicing
Done	Team reviewed class slides on requirements and use-cases before drafting Milestone 1

Check List	
Done	Team leads checked Milestone 1 document to quality, completeness, formatting and compliance with instructions before the submissions
Done	Team shared and discussed experience with GenAI tools among themselves
Done	GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone document etc.
Done	Team has chosen task management tool and invited Course instructor to project via ajsouza@sfsu.edu email
Done	Team reviewed non-functional requirements from the How to Start document and developed Milestone 1 consistently.
Done	Team lead ensured that all team members read the final M1 and agree and understand it before submission.